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FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2024

DIGITAL SIGNAL PROCESSING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	5 x2=10		
1.	a.	What is the need for frequency domain sampling?	L2	CO1	02
	b.	How does the Discrete Fourier Transform (DFT) relate to the Z-Transform?	L1	CO1	02
	c.	What is the main advantage of using the Radix-2 FFT algorithm for computing the DFT?	L2	CO2	02
	d.	How does the bilinear transformation help in avoiding aliasing?	L2	CO4	02
	e.	What is the purpose of using window functions like Hamming and Hanning in the design of FIR digital filters?	L2	CO6	02
		PART-B	**6 X 5=30**		
2.	a.	Compute the N – point DFT of the following signals. $(i)\ x(n) = \delta(n - n_0)\ ;\ 0 < n_0 < N$ $(ii)\ x(n) = an\ ;\ 0 \le n \le N-1$	L3	CO2	05
		OR			
	b.	Determine the 4 - point circular convolution of the sequences $x_1(n) = cos\left(\frac{2\pi n}{N}\right)$ and $x_2(n) = \sin\left(\frac{2\pi n}{N}\right)$; $0 \le n \le 3$ using the time – domain formula	L3	CO2	05
3.	a.	State and prove periodicity and linearity property of DFT.	L2	CO2	05
		OR			
	b.	Explain DFT as linear transformation	L2	CO1	05
4.	a.	Find the 4 point DFT of the sequence x(n)=(1,-1,1,-1).Also, using time shift property, find the DFT of the sequence, $y(n)=x((n-2))_4$	L3	CO2	05
		OR			
	b.	Compute the 4 point DFT of the sequence x(n)=(1,0,1,0).Also find y(n) if $Y(k)=X((k-2))_4$	L3	CO2	05
5.	a.	A system function of the normalized low pass filter is given below. $H(s) = \frac{s+1}{(s+1)^2+9}$.Determine H(z) using Impulse Invariance method. Assume T=1 sec.	L3	CO4	05
		OR			
	b.	Write note on Goertzel algorithm	L2	CO3	05

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6.	**a.**	Derive an expression for the order of Butterworth low pass filter in terms of passband frequency, stop band frequency, pass band gain and stopband attenuation.	L2	CO4	**05**
		OR			
	b.	Design a low pass Butterworth filter to meet the following frequency domain specifications: -1dB≤20log$\|H(j\Omega\|$≤0 for all Ω≤4 rad/sec 20log$\|H(J\Omega\| \leq -20$dB for all Ω≥8 rad/sec	L3	CO4	**05**
7.	**a.**	Find the order of the Chebyshev I filter to meet the following specifications Pass band ripple:≤2 dB Passband Edge:1 rad/sec Stop band attenuation:≥20 dB Stopband Edge:1.3 rad/sec	L3	CO5	**05**
		OR			
	b.	Find the order of an IIR Chebyshev digital filter that when used in the pre filter A/D-H(z)-D/A structure will satisfy the following specifications Low pass filter with -2 dB cutoff at 100Hz Stopband attenuation of 20 dB or greater at 500Hz And sampling rate of 4000 samples/sec	L3	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	State and prove **i.** Circular time shift and **ii.** Circular frequency shift property of DFT	L2	CO2	**10**
		OR			
	b.	Using DFT and IDFT find the linear convolution between the sequences x(n)=(1,0,1) and h(n)=(1,1)	L3	CO2	**10**
9.	**a.**	Develop radix – 2, DIT – FFT algorithm and write signal flow graph for $N = 8$.	L3	CO3	**10**
		OR			
	b.	Given x$(n) = (n + 1)$ for $0 \leq n \leq 7$. Find X(k) using DIT – FFT algorithm.	L3	CO3	**10**
10.	**a.**	Using overlap & save method, compute the output of an FIR filter with impulse response $h(n) = \{1, 2, 3\}$ and input $x(n) = \{2, -2, 8, -2, -2, -3, -2, 1, -1, 9, 1, 3\}$ use only 8 – point circular convolution in your approach	L3	CO2	**10**
		OR			

	b.	Using overlap –add method compute the output of an FIR filter with impulse response h(n)=(3,2,1,1) if the input x(n)=(1,2,3,3,2,1,-1.-2,-3,5,6,-1,5,6,-1,2,0,2,1)	L3	CO2	**10**
11.	**a.**	Draw the frequency response curve and write the transformation to convert the analog low pass prototype into practical analog low pass, high pass, bandpass and band stop filters with specified frequency.	L3	CO4	**10**
		OR			
	b.	A digital low pass filter is required to meet the following specifications 1.Monotonic passband and stop band 2.-3.01 dB at a cutoff frequency of 0.5π rad 3. Stop band attenuation of at least 15 dB at 0.75π rad. Find the system function H(z).	L3	CO4	**10**
12.	**a.**	Design an analog Chebyshev filter with the following specifications Passband ripple:1dB for $0 \leq \Omega < 10$ rad/sec Stop band attenuation:-60 dB for $\Omega \geq 50$rad/sec	L3	CO5	**10**
		OR			
	b.	An analog filter is given by $H_a(s) = \frac{1}{(s+3)(s+1)}$ with T=1 sec. Find H(z) Using Bilinear Transformation Method.	L3	CO5	**10**
13.	**a.**	Desired frequency response of the low pass filter is given by $H_d(e^{j\omega}) = \begin{cases} e^{-j3\omega}; & \lvert\ \rvert \leq 3\pi/4 \\ 0; & 3\pi/4 \leq \lvert\ \rvert \leq \pi \end{cases}$ Determine the frequency response of FIR filter if the Hamming window is used with N=7.	L3	CO6	**10**
		OR			
	b.	Determine the impulse response h(n) of a filter having desired frequency response $H_d(e^{j\omega}) = \begin{cases} e^{-j\ (N-1)/2} & for\ 0 \leq \lvert\ \rvert \leq \pi/2 \\ 0 & for\ \pi/2 \leq \lvert\ \rvert \leq \pi \end{cases}$ N=7.Use frequency sampling method	L3	CO6	**10**